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Greenhouse gas mitigation potential of solar photovoltaics: a highly variable indicator within the European Union



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Solar photovoltaic (PV) is a key technology for decarbonization. However, these systems cause an environmental impact along their life cycle. Greenhouse gas (GHG) emissions and mitigation from PV have been studied on a world level, with a yearly resolution, concluding that there are significant differences in decarbonization depending on where PV systems are deployed. This study explores the life-cycle GHG mitigation potential in the European Union countries with hourly resolution. The life-cycle GHG mitigation potential from PV can vary from 0.6 t CO₂e kWp⁻¹ in Sweden to 18.6 CO₂e kWp⁻¹ in Cyprus. Furthermore, a difference between calculating with an hourly and yearly resolution up to 49% is found. Despite this estimation being dependant on future decarbonization scenarios and other limitations, the variation between countries suggest that these results could be used as a decision-making tool to prioritize PV deployment in regions with higher mitigation potential.

The reduction of greenhouse gas emissions (GHG) is crucial to mitigate climate change¹. Renewable energies, including solar photovoltaic (PV), are key technologies to achieve the decarbonization goals¹. In 2024, PV generated 7% of the global electrical energy². Furthermore, PV capacity is growing, it is projected to triple by 2030³. China accounts for nearly 60% of global renewable capacity growth, with solar PV representing 80% of its expansion⁴. However, from a life-cycle perspective, PV is not an emission-free technology because during the material extraction, manufacturing, transport, installation, and end of life of these systems, GHG are emitted⁵. Nevertheless, in 2019, the global PV market mitigated more than 4 times the GHG produced⁶. A previous study projected that if the deployment is focused on carbon-intensive regions and the manufacturing is prioritized in Europe and North America, the GHG mitigation could almost double by 2060 compared with other projections⁶. This and other scientific studies have explored the difference that the deployment strategies make in mitigating GHG globally and advocate for strategic international coordination⁷.

Currently, some organisations involve several countries around the world to collaborate on renewable energy deployment, such as the Race to Zero campaign⁸, the International Energy Agency (IEA)⁹, and the International Renewable energy agency (IRENA)¹⁰. On a continental level, the EU represents a successful collaboration framework with proven results, integrated policies, and substantial financial commitments. In the EU, renewable energy has reached almost a 48% share of the total electrical energy generation during 2024². REPower EU mobilized close to \$300 billion for renewables and reached almost 338 GW of new solar energy

capacity installed¹¹. Therefore, the EU renewable energy framework could benefit from a deployment strategy that achieves faster decarbonization, as shown by the world-level studies⁶. However, a review on PV power deployment in central European countries shows that even with a variety of policy measures, a stronger influence is given by a set of national rules and financial support¹².

The GHG mitigation potential from PV deployment has been assessed in previous research as the product of the PV energy generation potential and the grid carbon intensity⁷. The PV generation is influenced by several environmental factors that change depending on the location. Irradiance has been determined as the main factor influencing PV generation¹³. However, other factors such as temperature, dust and snow also affect the performance of PV systems¹³. Furthermore, the degradation of the PV modules, which causes a reduction of the PV generation with time, also varies depending on the environmental factors of a given location¹⁴. Regarding the lifecycle grid carbon intensity, it is calculated as the total lifecycle GHG emissions divided by the total electrical energy generated in each area of the electrical grid¹⁵.

The GHG mitigation potential of PV allocation strategies has been studied previously, on a global and national scale. One study evaluated 1 kWp of three PV technologies to determine the best manufacturer-installation scenario for providing the highest reduction in GHG, examining 141 locations worldwide⁷. In a more recent publication, the total worldwide GHG emissions and mitigation of PV deployment was evaluated from 2009 to 2019 and projected from 2020 to 2060⁶. Furthermore, the global warming

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mitigation for rooftop was analyzed using worldwide data with a 1-km resolution¹⁶. These three studies used average grid emissions with a yearly resolution and acknowledged their limitations due to the daily and seasonal variability. The variation between using hourly and yearly resolution for GHG mitigation assessment was studied for a case in Spain, a hypothetical roof installation of 1 kWp was analyzed and the results for the hourly resolution GHG mitigation were 8% lower than the yearly one¹⁷. Besides these findings in Spain, a study on hourly resolution assessment of the GHG mitigation potential of PV deployment for multiple countries was not found during the literature review.

The analysis presented here is based on the premise that coordination for deployment could contribute to decarbonization goals⁶, and the strong collaboration around renewable energy within the EU¹¹. The aim of this analysis is to quantify the life-cycle GHG mitigation potential of a 1 kWp PV system for each of the 27 EU countries, with an hourly resolution for the sake of more accurate results. The life-cycle GHG mitigation potential represents the cumulative lifetime GHG mitigation by summing the hourly mitigation (PV power generation times grid's carbon intensity) across all operating hours. (for more details, see the Method section).

Results

Life-cycle GHG mitigation potential

The results show a significant variation of life-cycle GHG mitigation potential within the analyzed locations, with the PV system in Cyprus having 31 times that of the system in Sweden due to the high solar irradiation and carbon-intensive electricity grid. The analysis of the relation of grid's carbon intensity and PV generation with life-cycle GHG mitigation potential shows grid's carbon intensity as the dominant factor for the life-cycle GHG mitigation potential. Moreover, the results are higher than the life-cycle GHG emissions of three PV systems obtained in a previous study¹⁸ for all cases except in Sweden and France, where one system (with simulated manufacturing in China) has higher life-cycle GHG emissions compared with the mitigation in these two countries. In addition to that, three different variations of the life-cycle GHG mitigation results are tested: (1) the difference between quantifying with hourly and yearly resolution, (2) projecting the results along the PV's life-cycle with future grid decarbonization, and (3) the possible influence of negative electrical energy prices on the mitigation potential.

The main contribution of this study lies in the hourly resolution of the analyses, which increases the accuracy of the results compared with previous studies with similar objectives. From the analyses performed, the highest variation found for exponential decarbonization, the values can be up to 70% lower compared to a quantification that does not consider this factor. Despite the possible variations, the results of life-cycle GHG mitigation potential from PV could be used to coordinate its deployment within the EU to foster a faster decarbonization, leveraged by the already existing strong collaboration among these countries.

The life-cycle GHG mitigation potential is quantified for the 27 countries of the European Union, using average hourly grid carbon intensity and the energy generation from a simulated 1 kWp PV system. The results, in Fig. 1, show a significant variation between countries. The highest result is in Cyprus, where the mitigation potential is 18.6 t CO₂e kWp⁻¹, and the lowest one is in Sweden, with 0.6 t CO₂e kWp⁻¹. The remaining results for the 25 countries are between these two extremes, with an average of 6.3 t CO₂e kWp⁻¹.

The locations in the North (latitudes higher than a 53° latitude) show a relatively low GHG mitigation potential due to the low solar irradiation and relatively low-carbon grids. The exception is Ireland with 9.6 t CO₂e kWp⁻¹, caused by a relatively high carbon grid with a high reliance on gas turbines¹⁹. In the lower end are the Nordic countries with values ranging from 0.6 to 2.4 t CO₂e kWp⁻¹. The simple average for the Northern Europe group of countries (Denmark, Estonia, Finland, Ireland, Lithuania and Sweden) is 3.8 t CO₂e kWp⁻¹, lower than the 5.8 t CO₂e kWp⁻¹ for mid-latitude locations.

The results for mid-latitude locations (latitudes between 48° and 53°) range from 0.8 t CO₂e kWp⁻¹ in France due to a high share of nuclear energy²⁰ and 14.5 t CO₂e kWp⁻¹ in Poland, the second highest in the EU

countries due to a carbon-intensive grid. The result for Germany is 6.4 t CO₂e kWp⁻¹, close to the average for the 27 EU countries (6.3 t CO₂e kWp⁻¹). The simple average for this group of countries is 5.8 t CO₂e kWp⁻¹, around 40% lower compared to the average of the Southern European countries, 8.2 t CO₂e kWp⁻¹.

Southern European locations (latitudes lower than 48°) have generally a higher PV generation, which influences the results of the life-cycle GHG mitigation potential. Portugal and Spain are on the west, and both have the lowest life-cycle GHG mitigation potential in this region with 3.4 t CO₂e kWp⁻¹ and 3.5 t CO₂e kWp⁻¹, respectively; despite high irradiation, and due to a relatively low-carbon grid compared to the rest of the countries in this group. The islands have the highest results, Malta with 13.8 t CO₂e kWp⁻¹ and Cyprus (the only country in the EU that is isolated from other grids²¹) with a life-cycle GHG mitigation potential of 18.6 t CO₂e kWp⁻¹, due to the combination of high PV power generation and a carbon-intensive grid. The rest of the southern European group is between 5 and 9 t CO₂e kWp⁻¹.

Relating mitigation potential to PV and grid factors

To illustrate the influence of the grid's carbon intensity and the PV generation on the life-cycle GHG mitigation potential, these variables are plotted in Fig. 2 with the PV penetration in 2024 from Ember's European Electricity Review 2025²². The analysis of these variables indicates that the grid's carbon intensity is the predominant factor for the life-cycle GHG mitigation potential. The life-cycle GHG mitigation potential varies more along the carbon intensity (Y axis) compared with the annual PV generation (X axis) because the relative variation in the carbon intensity, encompassing more than 30-fold range (~700/22) is higher than the relative variation for the PV generation which is less than a 2-fold range (~1670/930). Figure 2 also shows PV penetration, this variable ranges from 1% in Finland to 24% in Hungary²². Cyprus, having the highest life cycle GHG mitigation potential has one of the highest PV penetration rates with almost 20%²².

GHG mitigation potential vs. PV system emissions

When the results previously shown are compared with the life-cycle GHG emissions from a PV system (including raw material extraction, manufacturing, transport, installation, and end of life), a conclusion about the net impact can be drawn. If the life-cycle GHG mitigation potential is higher than the life-cycle GHG emissions of a given PV system, a potential benefit towards decarbonization can be assumed. The life-cycle GHG mitigation potential compared with the emissions from three PV systems manufactured in China, Germany, and Norway (obtained from a recent study¹⁸) is presented in Fig. 3.

Comparing the life-cycle GHG mitigation potential with the life-cycle GHG emissions of three PV systems (modules with manufacturing simulated using the electricity mix of China, Germany, and Norway¹⁸) shows that the three PV systems have a theoretical decarbonization benefit in most locations. There are only two exceptions: the PV system from China has a global warming potential of 1.1 t CO₂e kWp⁻¹, 75% and 42% higher than the GHG mitigation potential of Sweden and France, respectively. For the remaining 25 countries, any of the three PV systems could theoretically contribute to the grid decarbonization, with up to 40 times more life-cycle GHG mitigation compared to the corresponding emissions of the PV system (Norwegian modules installed in Cyprus).

Hourly vs. yearly resolution in GHG mitigation potential

The results of quantifying the life-cycle GHG mitigation potential with yearly and hourly resolution are shown in Fig. 4. The results are similar for most countries, with two exceptions. The estimation with a yearly resolution is generally higher compared to a more accurate hourly resolution quantification, up to 49% higher for Cyprus. The two locations where the yearly quantification method gives higher results than the hourly obtained ones are Sweden and Ireland, with variations of -15% and -6% respectively. These differences can be explained by the daily and seasonal variation of both the solar energy generation from the 1 kWp system and the grid energy mix in the different countries.

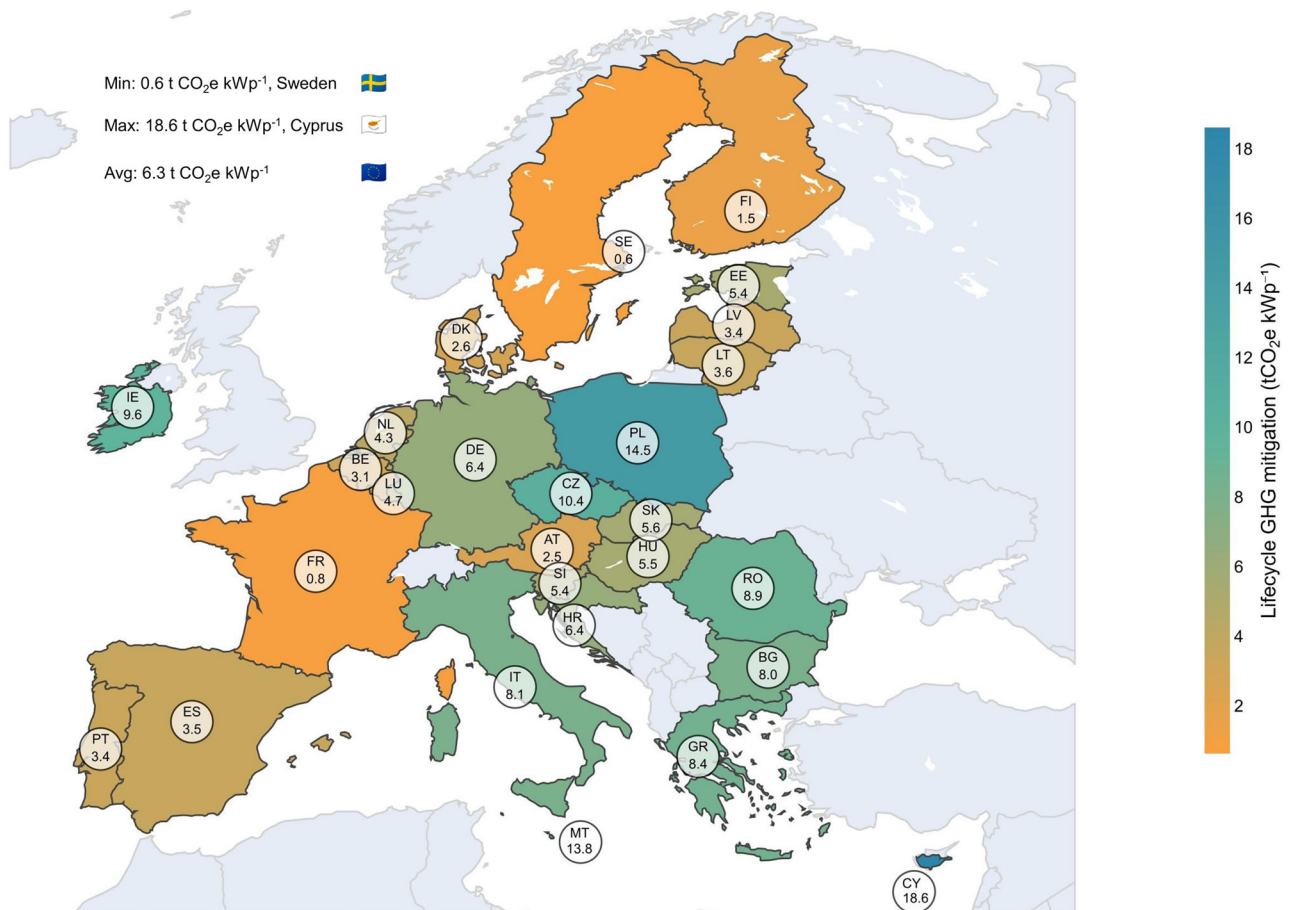


Fig. 1 | The life-cycle GHG mitigation potential in European Union countries in ton CO₂e per unit of installed kWp PV.

Fig. 2 | Life-cycle GHG mitigation potential is plotted as bubbles with the average PV generation (X axis) and the 2024 yearly average life-cycle carbon intensity (Y axis). Bubble fill level indicates the proportion of total electricity generation from solar PV in 2024.

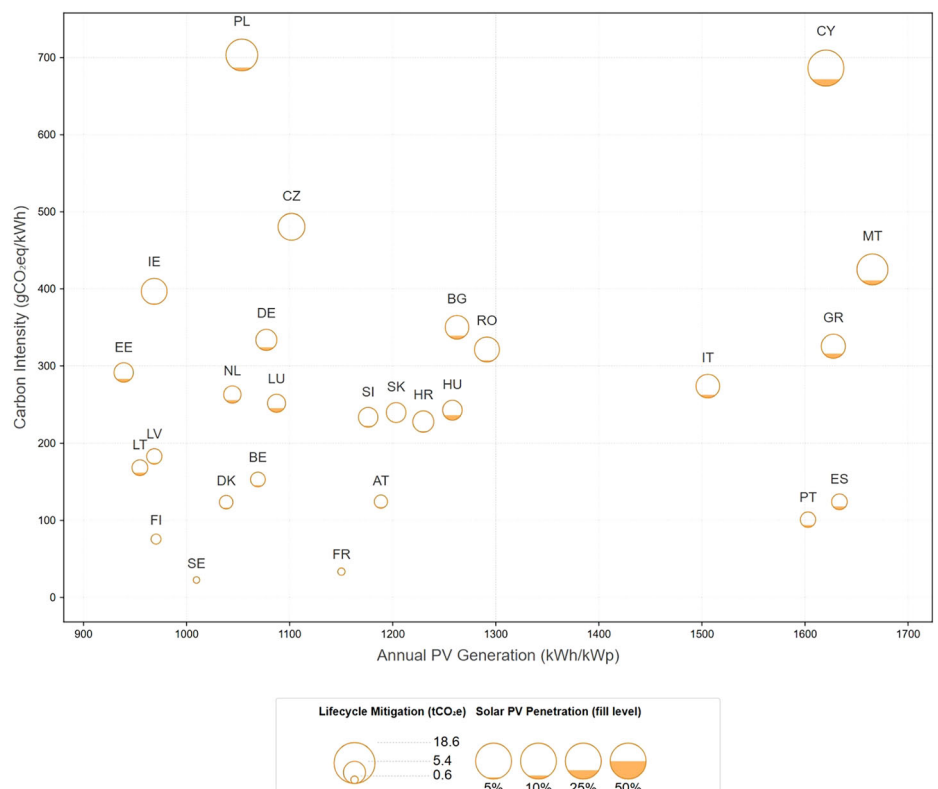


Fig. 3 | A comparison between life-cycle GHG mitigation potential represented by the gray bars (from this study), and the LCA GHG emissions from PV systems using the electricity mixes of China, Germany, and Norway from a previous study¹⁸, represented by vertical dashed lines.

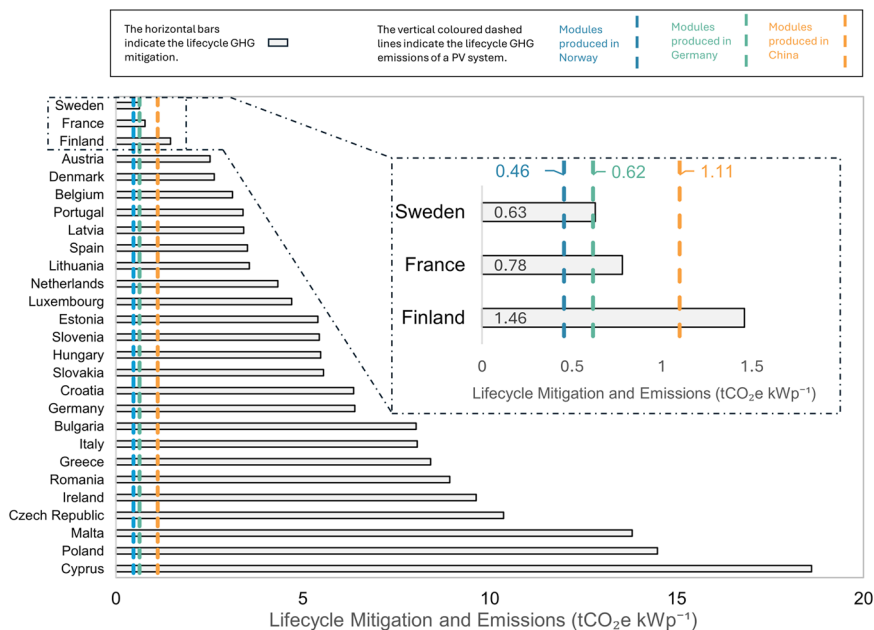
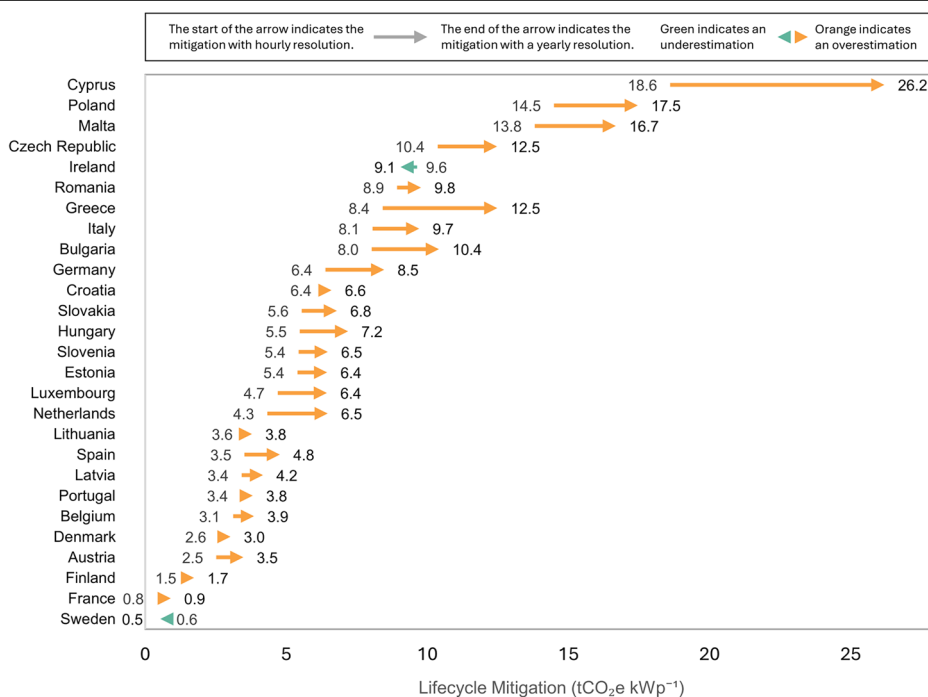


Fig. 4 | The variation of the life-cycle GHG mitigation potential between quantification with hourly resolution and yearly resolution for EU countries, in ton CO₂e per unit of installed kWp PV.



The variation while using a yearly resolution (instead of hourly) to quantify the life-cycle GHG mitigation potential is generally higher for countries with a high solar energy penetration in their grids. This caused the grid carbon intensity to be lower when the PV system generation was high, making the periods of high PV generation account for less mitigation than if simply average yearly carbon intensities were used. This effect causes a higher variation in countries like the Netherlands (49%), Greece (48%), and Cyprus (41%). The average resolution variation for the 25 countries where the yearly estimation is higher than the hourly is 24%, considerably higher than the average for the opposite case, found in Sweden and Ireland (10%).

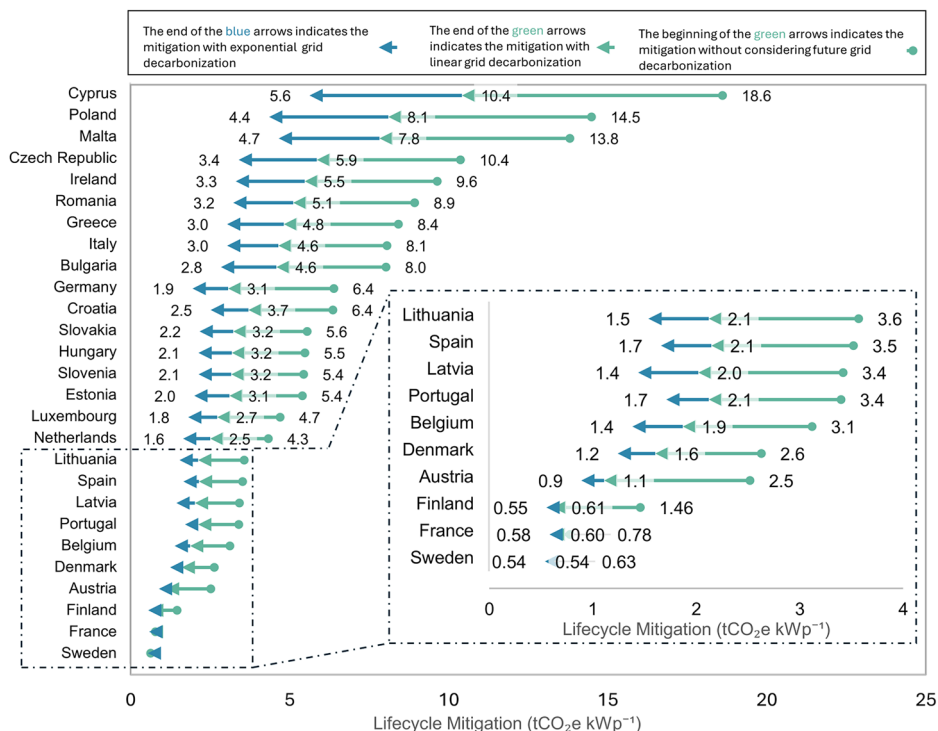
The two locations where the quantification with yearly average showed a lower life-cycle GHG mitigation potential are where the periods of high PV

production have a higher or similar carbon intensity than other times during the year. This causes that periods with high PV generation account for more mitigation than if average yearly carbon intensities were used. This is the case for Sweden, where PV power generation has a higher lifecycle emissions factor than the average of the remaining energy sources in the grid, and in Ireland, where the patterns of wind power generation cause a slightly higher average carbon intensity during periods with high PV energy generation.

GHG mitigation potential under grid decarbonization

The life-cycle GHG mitigation potential changes depending on the scenario for future grid decarbonization used in the assessment of the 1 kWp PV system 25 years into the future, starting from 2024. The results shown previously do not assume any grid decarbonization; the hourly carbon intensity

Fig. 5 | Variation of theoretical lifecycle GHG emissions mitigation by assuming no decarbonization, linear, and exponential decarbonization.



values from 2024 are assumed as the same 25 years into the future. Figure 5 shows the variation of the previous results with linear and exponential decarbonization paths based on the national net zero targets (as explained in the methods section). The decarbonization scenarios show lower GHG mitigation potential for the PV system, compared with the no-decarbonization assumption.

The exponential decarbonization assumption (a faster reduction of grid carbon intensity at the beginning) caused a higher reduction in the lifecycle GHG emissions mitigation compared to a linear (steady) decarbonization path. In general, a higher GHG mitigation potential causes a higher variation between the three decarbonization projections. For example, in Cyprus the variation is -40% and -70%, for the linear and exponential decarbonization, respectively. On the other hand, in France the results showed a variation of -23% for linear and -25% for exponential decarbonization. The variation in the results for both countries is significantly different, despite both having the same 2050 target year.

For countries with similar life-cycle GHG mitigation potential, the highest variation occurs when the net-zero targets are closer. For example, Finland and France have a lifecycle GHG mitigation without a decarbonization projection of 1.46 t CO₂e kWp⁻¹ and 0.78 t CO₂e kWp⁻¹ and net-zero targets in 2035 and 2050, respectively. The earlier target for Finland caused a lower lifecycle GHG mitigation than France when exponential decarbonization is projected, 0.55 t CO₂e kWp⁻¹ for Finland and 0.58 t CO₂e kWp⁻¹ for France. Other results more sensitive to decarbonization projection due to earlier targets are Germany and Sweden, with a net-zero target in 2045.

GHG mitigation potential considering electricity prices

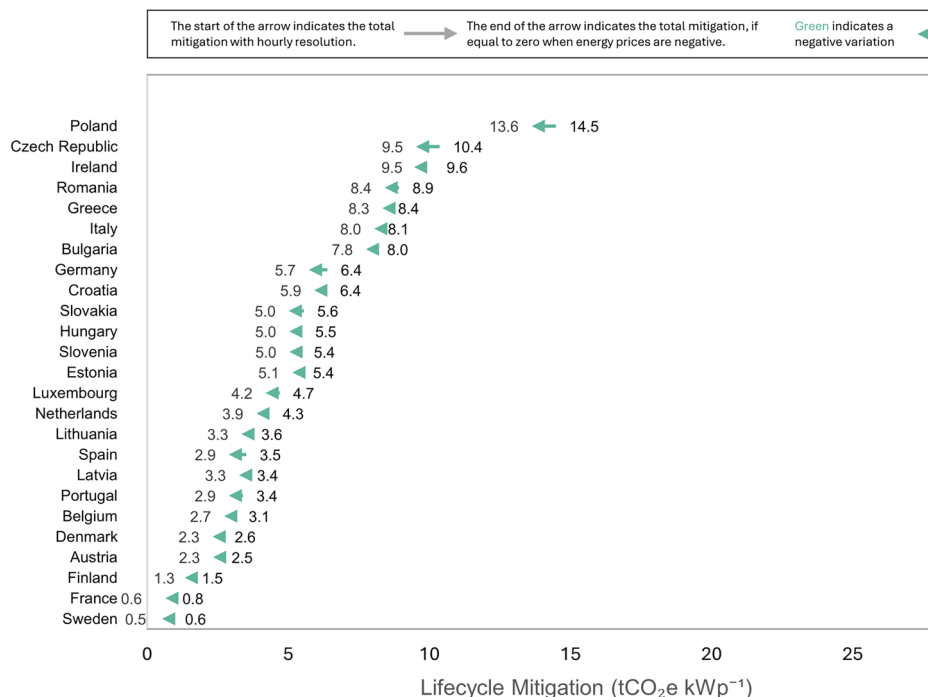
If negative electrical energy prices are considered as a proxy for curtailment (non-decarbonization benefit) the life-cycle GHG mitigation potential is reduced, as shown in Fig. 6. This analysis is performed for all EU countries except Cyprus and Malta, due to the lack of publicly available hourly electricity prices. For the 25 remaining countries, the average reduction in lifecycle GHG mitigation is 8%, the highest reduction is 18% for Spain, and the lowest is 0% for Italy. The reduction in Spain is because there were 247 h of negative prices during 2024, while in Italy, there were no negative prices due to the regulation².

Discussion

This study explores the life-cycle GHG mitigation potential of a 1 kWp PV system, optimally oriented, installed in locations in each of the 27 EU countries. As in other studies, significant differences between countries were found, with high emissions and high irradiation countries having the highest mitigation⁷. If the results from this study are analyzed next to the PV added capacity during 2023³, there is no obvious evidence of installation being prioritized in areas with high mitigation potential. This suggests that the current mechanisms to prioritize PV where it brings the biggest benefit, if any, might not be effective. Furthermore, PV deployment is driven by explicit renewable energy sources support policy, profitability of investment and certainty about the government's long-term deployment intentions²³, which are given by EU level policies and more importantly a set of national rules¹². Therefore, these results could be used to inform a policy within the EU that prioritizes the countries where PV brings higher decarbonization benefits, as other studies have advocated for in a Worldwide level⁷.

The variation analyses confirm findings from previous studies with some differences. The variation between life-cycle GHG mitigation potential quantified with yearly average and hourly average emissions for Spain found in this study is 36% while for a previous reference it is 8%¹⁷. This variation could be explained by the energy mixes used; the previous study had 2021 mix data¹⁷, when the grid had a lower installed solar capacity. This study uses 2024 mix data from a grid with around double solar capacity, from 16 GWp (2021) to 31 GWp (2023)²⁴. Also, the difference between estimating the mitigation potential with a constant future emissions factor and with decarbonization projections was found to be considerable. Assuming future decarbonization causes a variation of the life-cycle GHG mitigation potential that ranges from 14% to 70% lower than that assuming constant grid GHG. A previous study found a variation of around 27% between not assuming decarbonization and actual decarbonization during the period between 2009 to 2019⁶. Regarding the negative electrical energy prices analysis, the results are higher than the publicly available curtailment rates with an average 8% reduction in the GHG mitigation potential, while official figures from IEA for curtailment are between 1.5% and 4%²⁵. The availability of detailed, hourly curtailment data is a limitation for this analysis.

Fig. 6 | The variation of life-cycle GHG mitigation potential by assuming PV generation curtailment when prices are negative.



The conclusions from each of the analysis presented previously can serve as part of a broader decision support framework for practitioners and policy makers that considers the limitations of this study. For example, the regional differences in the lifecycle mitigation potential of PV systems can guide decision making on to where to deploy PV. However, this could imply additional challenges due to higher PV penetration like cannibalization effect (increasing renewable technologies' penetration undermines their own value²⁶) and requirements for flexibility to avoid grid instability²⁷. For example, Cyprus (a special case) has the highest mitigation potential, yet also already has a relatively high PV penetration²², as does Greece²². Another finding from this study is the increase in accuracy that hourly (instead of yearly) resolution provides in the lifecycle GHG mitigation potential. Again, Cyprus' results can serve as an example of the difference caused by the resolution of the calculation, because its relatively high PV penetration causes an overestimation (when using a yearly grid carbon intensity) of the GHG mitigation potential during high PV generation, typically lower carbon intensity intervals. Another contribution is the difference that estimating the mitigation potential with and without decarbonization makes, which depends on a projection that is inherently uncertain. Acknowledging and quantifying this uncertainty is also a practice that can help in understanding how today's decisions might develop along the lifespan of a PV system being deployed in the present.

In this study, 27 PV systems are simulated using PVGIS with a specific configuration and considering only crystalline silicon module technology and PVGIS-SARAH2 surface radiation data record²⁸. Therefore, only one point for every country is simulated, which might not comprehensively cover the performance diversity within each country due to technology, climate, and grid conditions. For example, temperature effects that are not fully captured can cause a variation of up to 10%, and this uncertainty increases if climate change is considered¹³. Furthermore, other simplifications made in the model, such as dust²⁹, humidity, snow, albedo and atmospheric factors can affect the PV performance¹³. Moreover, all these factors, particularly the temperature can also affect the degradation of the PV modules, which in turn also affects the PV power generation over time¹⁴. Therefore, a future study with a higher geographic resolution^{16,30}, more PV technologies⁷ and more accurate PV generation modeling can give more detailed insights into the GHG mitigation potential of PV in EU countries.

Another set of limitations arises from the assumptions around the life-cycle GHG emissions and mitigation. The analysis was performed with average hourly and yearly emissions, as if the energy generated from the PV would mitigate the GHG from an average unit of energy from the grid during that time. However, the mitigation can be related to the change in the energy system, usually called marginal energy, usually assumed to be more carbon-intensive due to the reduction of controllable, often fossil fuel energy sources, such as a gas turbine¹⁷. Furthermore, the PV energy could be displacing energy outside the electrical grid, including electric vehicles or electrified industry. Additionally, the growing adoption of battery storage alongside PV systems can shift the time of grid displacement, potentially altering the carbon intensity of the displaced electricity and thus the mitigation potential; this effect is not captured in the present analysis but warrants consideration in future studies. An application-based GHG mitigation for PV could be explored in future research in a similar way that was done for hydrogen³¹. However, the highly complex and dynamic nature of energy systems would cause estimating the marginal GHG emissions to carry more uncertainties than using a more conservative average quantification. Similarly, the embodied GHG taken from the literature¹⁸ are also subjected to change. For example, the grid emissions used for the manufacturing from China are potentially lowering the future due to the high rate of PV deployment⁴.

Despite the limitations and uncertainties in this study, the differences in life-cycle GHG mitigation of PV between countries are up to 10 to 30 times, if the results from Sweden and Cyprus are compared. The differences between countries have been found by more studies on a wider context and different methods⁶. Therefore, the results from this analysis could inform mechanisms for a coordinated deployment of PV, leveraged by the existing collaboration within the EU to achieve a faster decarbonization. These collaboration mechanisms could spread worldwide and help to achieve a more rapid climate change mitigation from PV systems globally.

Methods

Life-cycle assessment framework and study design

The framework for this study is lifecycle assessment (LCA) according to ISO 14044³². The main goal of the LCA is to determine the change environmental impact due to the installation of 1 kWp of PV in the EU countries. The system function is to provide the same electrical energy from the grid as the generated from a PV system across a lifespan of

Table 1 | Location and orientation of the PV systems simulated

Country code	Country name	Latitude	Longitude	Optimal tilt (degrees)	Optimal azimuth (degrees)
AT	Austria	48.21	16.37	39	−2
BE	Belgium	50.85	4.35	39	−3
BG	Bulgaria	42.70	23.32	36	−7
CY	Cyprus	35.19	33.38	32	−3
CZ	Czech Republic	50.08	14.44	39	−5
DE	Germany	52.52	13.41	40	−5
DK	Denmark	55.68	12.57	42	1
EE	Estonia	59.44	24.75	43	2
ES	Spain	40.42	−3.70	37	−4
FI	Finland	60.17	24.94	44	2
FR	France	48.86	2.35	38	−3
GR	Greece	37.98	23.73	33	−1
HR	Croatia	45.82	15.98	36	−3
HU	Hungary	47.50	19.04	39	−1
IE	Ireland	53.35	−6.26	41	−3
IT	Italy	41.90	12.50	37	−3
LT	Lithuania	54.69	25.28	40	−2
LU	Luxembourg	49.61	6.13	37	−2
LV	Latvia	56.95	24.11	42	−1
MT	Malta	35.90	14.51	32	3
NL	Netherlands	52.37	4.90	40	1
PL	Poland	52.23	21.01	39	−3
PT	Portugal	38.72	−9.14	33	4
RO	Romania	44.43	26.10	35	2
SE	Sweden	59.33	18.07	44	−2
SI	Slovenia	46.06	14.51	35	7
SK	Slovakia	48.15	17.11	38	−2

25 years, hence the functional unit is a 1 kWp PV system with optimized orientation to maximize energy generation, operating for 25 years. The lifecycle scope is cradle to grave (raw material extraction, manufacturing, transport and end of life) and the geographical scope is one specific location per country in the EU. The coordinates of the locations and the optimal orientation and tilt (according to PVGIS) are shown in Table 1.

The lifecycle inventory includes the electrical AC energy from the grid, equivalent to the generated by a 1 kWp of PV across 25 years with 0.5% yearly degradation. The environmental impact assessment is done using electricity maps life-cycle emissions data for every hour³³, which is obtained from the weighted average per energy source in the corresponding price zone using IPCC and UNECE life-cycle emission factor plus energy transmission from other price zones³⁴. The environmental impact category used is the global warming potential (GWP), it quantifies the potential warming effects of GHG relative to carbon dioxide, one hundred years in the future³⁵. The results of this LCA are referred to as “life-cycle GHG mitigation potential” within this text. The analysis of the results is done by comparing the 27 cases with each other plus four additional analyses: (1) comparative analysis of the results with the GWP of three PV systems with different manufacturing scenarios taken from the literature¹⁸, (2) variation analysis of the results with yearly resolution, (3) variation of the results with decarbonization scenarios, and (4) variation considering curtailment when prices are negative.

Figure 7 provides an overview of the methodology, which begins with data from 27 locations and concludes with the analyses presented in the previous figures. In the diagram, white shapes represent the analytical stages (which use referenced, simulated, and transformed data) while the colored shapes represent the main results plotted in each figure.

Average and life-cycle hourly power generation

To make the entire analysis, one location from each of the EU countries is selected³⁶; the coordinates for the sites are shown in Table 1. These locations are used to simulate the hourly PV power generation from 2015 to 2020, for a 1 kWp system with optimal orientation using PVGIS²⁸. From the six-year power generation data, the hourly power generation for an average year is obtained as shown:

$$P_h^{mean} = \frac{1}{y_{p,n} - y_{p,0}} \sum_{y=y_{p,0}}^{y_{p,n}} P_{h,y} \quad (1)$$

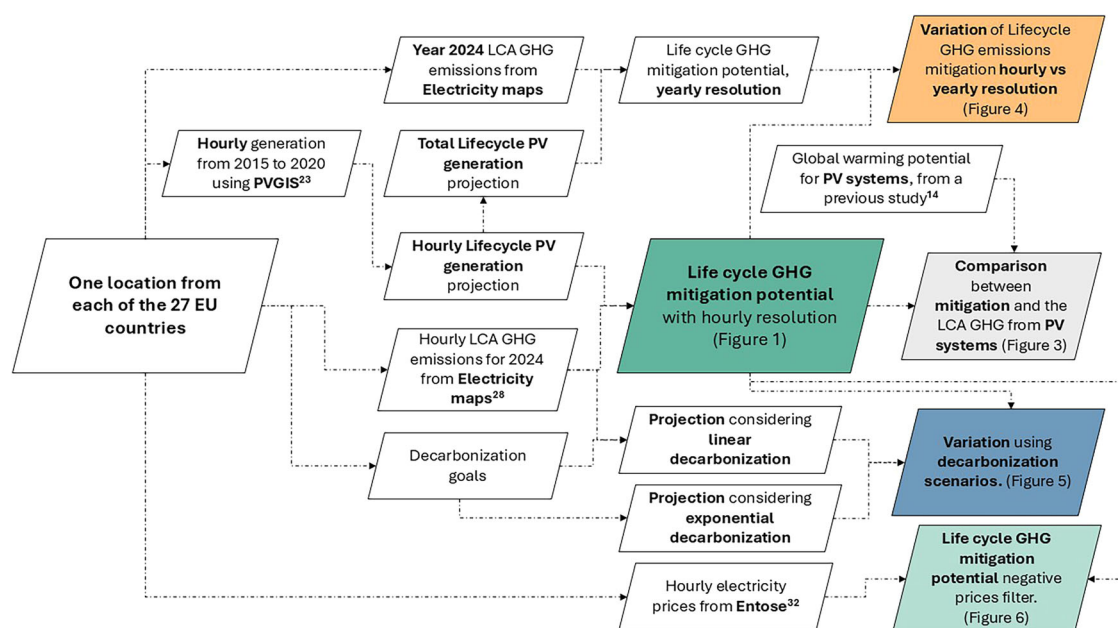


Fig. 7 | The flowchart shows an overview of the methods used to obtain the results related to the life cycle GHG mitigation potential of PV systems in EU countries.

where $P_{h,y}$ represents the power generated during each hour from 2015 to 2020 (expressed as $y_{p,0}$ and $y_{p,n}$ respectively), y represents a given year within this range and P_h^{mean} is the average hourly power generation across that period. From the average year, the lifecycle PV energy generation on an hourly basis is calculated for 25 years using a degradation of 0.5%, according to the following equation:

$$E_h^{lc} = \sum_{y=y_0}^{y_n} (1 + d)^{y-y_0} \cdot P_h^{mean} \quad (2)$$

where d is the degradation and E_h^{lc} is the lifecycle from y_0 to y_n (2024–2048) energy generated from the PV system during each hour on an average year. Having lifecycle hourly energy generation data facilitates further analysis.

Life-cycle GHG mitigation potential

To get the life-cycle GHG mitigation potential, the life-cycle energy generation data is multiplied by the hourly LCA GHG emissions for 2024, obtained for the corresponding grid area from electricity maps³³, and summed as follows:

$$A_{lc}^{hr} = \sum_{h=1}^{h_n} E_h^{lc} \cdot G_h \quad (3)$$

here G_h is the LCA GHG for a given hour (h) during 2024 from $h = 1$ to h_n (the last hour of the year), and A_{lc}^{hr} represents the life-cycle GHG mitigation potential calculated with hourly resolution. These are the main results, and they are plotted in Fig. 1. The following analysis is a comparison between the lifecycle GHG mitigation potential and lifecycle GHG emissions from a previous study for a PV system with three manufacturing scenarios: Norway, Germany, and China¹⁸. The remaining three analyses are to explore possible variations of the assumptions made.

Variation analyses on life-cycle GHG mitigation potential

To illustrate possible variation in the results, due to different assumptions, three variation analyses were done: (1) Hourly and yearly resolution, (2) Negative price effects, and (3) Decarbonization scenarios. To analyze the difference between calculating with yearly resolution and hourly resolution, the theoretical mitigation with yearly resolution was obtained by multiplying the total energy generation (obtained from the sum of the life-cycle hourly generation) by the yearly 2024 LCA carbon intensity, as shown in the following equation:

$$A_{lc}^{yr} = G_y \cdot \sum_{h=1}^{h_n} E_h^{lc} \quad (4)$$

where G_y is the yearly greenhouse emissions of the given price area, and A_{lc}^{yr} represents the life-cycle GHG mitigation potential calculated with yearly resolution. The negative price analysis is used as a proxy to estimate how much mitigation potential is lost due to curtailment. For this purpose, the GHG mitigation potential in a given hour is considered zero when the electricity price, for 2024 obtained from ENTSO-E³⁷, is negative and not affected when the price is zero or positive. After these changes, the hourly mitigation potential is summed, as follows:

$$A_{lc}^{hr,v} = \sum_{h=1}^{h_n} \begin{cases} E_h^{lc} \cdot G_h & \text{if } V_h > 0 \\ 0 & \text{if } V_h \leq 0 \end{cases} \quad (5)$$

here V_h represents the electricity spot price for a given hour in 2024, and $A_{lc}^{hr,v}$ is the life-cycle GHG mitigation potential calculated with hourly resolution, assuming no mitigation when prices are negative. The last variation analysis is done to explore how two different decarbonization paths could affect the results.

GHG mitigation potential under grid decarbonization

To analyze decarbonization scenarios, three additional assumptions are defined: (1) the decarbonization objective year, (2) the decarbonization path, and (3) the lifecycle GHG emission reached in the zero scenario. The decarbonization objective year is obtained from publicly available country or EU level goals. The most common goal year is 2050; however, Finland (2035)³⁸, Austria (2040)³⁹, Sweden (2045)⁴⁰, and Germany (2045)⁴¹ are exceptions. For the decarbonization path, two functions are used: linear and exponential decarbonization. Finally, for the minimum value achievable in the grid is taken from the report “Energy Pathways to 2050”, published by the French transmission system operator⁴². The value taken from this report is 16.3 g CO₂e kWh⁻¹, obtained from M1 scenario with focus on 100% renewable energy⁴², (10.5 Mt CO₂e/645 TWh). The defined assumptions are combined to obtain the lifecycle GHG emissions mitigation under linear and exponential decarbonization scenarios as shown in Eqs. (7) and (9), respectively. Equation (6) is used to obtain the linear decarbonization fraction for a given year, as a previous step to get life-cycle GHG emissions mitigation with linear decarbonization, and it is defined as follows:

$$F_y^{lin} = 1 + \left[\left(\frac{y - y_0}{y_g - y_0} \right) \cdot \left(\frac{G_{zero}}{G_y} - 1 \right) \right] \quad (6)$$

where G_{zero} is the zero emissions floor (16.3 gCO₂e kWh⁻¹), y_g is the decarbonization goal year, G_y is the LCA GHG from the grid for 2024, and F_y^{lin} is the fraction of the grid’s LCA GHG from 2024, for a in year y , under a linear decarbonization scenario. The result from Eq. (6) serves as input for the following equation:

$$A_{lc}^{lin,hr} = \sum_{y=y_0}^{y_n} \left[(1 + d)^{y-y_0} \cdot F_y^{lin} \cdot \sum_{h=1}^{h_n} (P_h^{mean} \cdot G_h) \right] \quad (7)$$

Here the result is $A_{lc}^{lin,hr}$, representing the life-cycle GHG mitigation potential under a linear decarbonization scenario. The life-cycle GHG mitigation potential with exponential decarbonization is calculated using the following two equations: is used to get the exponential decarbonization fraction, defined as follows:

$$F_y^{exp} = \left(\frac{G_{zero}}{G_y} \right)^{\left(\frac{y-y_0}{y_g-y_0} \right)} \quad (8)$$

where the input variables are the same as for Eq. (6), and the result is the fraction of the lifecycle GHG emissions from the grid compared to that from 2024, for a given year, under an exponential decarbonization scenario expressed as F_y^{exp} . The result obtained from Eq. (8) is used as input for Eq. (9), which is defined as:

$$A_{lc}^{exp,hr} = \sum_{y=y_0}^{y_n} \left[(1 + d)^{y-y_0} \cdot F_y^{exp} \cdot \sum_{h=1}^{h_n} (P_h^{mean} \cdot G_h) \right] \quad (9)$$

where $A_{lc}^{exp,hr}$ is the life-cycle GHG mitigation potential under an exponential decarbonization scenario.

Data availability

The data will be made public in a repository (to be created) after publication.

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